MILITARY SPECIFICATION
CARTRIDGE, 40MM, HEDP, M433
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Cartridge designated as Cartridge, 40MM HEDP, M433.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Method of
MIL-P-20444	-	Primer M42
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-650	-	Explosive, Sampling, Inspection Testing

DRAWINGS (See 6.10)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

8862362	-	Box, Fiber, Packing, Ammunition for Cartridge, 40MM, M397, M397E1 and M433
8882363	-	Box, Wirebound, Packing, Ammunition for Cartridge, 40MM, M397, M397E1, M433 and M407A2
8835104	-	Box, Wirebound, Packing, Ammunition for Cartridge 40MM
8835105	-	Box, Fiber, Packing, Ammunition for Cartridge 40MM
8886371	-	Cartridge, 40MM, HEDP, M433
9276547	-	RDX Blend

INSPECTION EQUIPMENT DRAWINGS

8827889	-	Chamber, Gage
8827890	-	Alignment Check
8827895	-	Limit Checks

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the data of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	-	Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg.) 8886371, all associated drawings, and all requirements specified in applicable specifications.

The following tests apply.

a. X-ray examination of ballistic samples.
b. Production control check on loaded M118 cartridge cases.
c. X-ray examination of fuze assemblies prior to assembling to cup and liner assembly.
d. Security of crimp of base plug on cartridge case assembly.
e. Pull test of spitback assembly from fuze assembly.
f. Specific gravity of the charge of cup and liner loading assembly.
g. Torque of crimp of fuze assembly on projectile.
h. Security of projectile.
i. Air pressure test of cartridge.
j. Primer functioning.
k. Propellant weight of cartridge case.
l. Propellant weight of cartridge case. (Alternate method)
m. Check test for deterioration of primer.
n. Ballistic functioning.
o. Heat seal test of packing seals.
p. Suitability test plan (informational).
q. Holding fixture vent check.

3.3 Contamination and distortion of Lexan parts. Extreme care should be taken throughout all phases of manufacture to avoid contacting "Lexan" parts with certain solvent materials and vapors which may attack the "Lexan" material. Specific information on "Lexan" destructive solvents may be obtained from the General Electic Co., Chemical and Metallurigical Division, Pittsfield, Mass.

3.4 Loading composition. The loading composition at the loading station at the time of loading the cup and liner shall comply with the requirements specified on Drawing (Dwg.) 9276547, and with all requirements specified in applicable specifications.

3.5 Loading of high explosive.

3.5.1 The fuze seating surface shall be free of caked explosive. A chemical cleaner, if used, shall not contaminate the explosive.

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3.6 Primer functioning. The primer shall function without evidence of squibbing or hangfire. Paragraph 6.6 or MIL-P-20444B, Amendment 3, contains definition of functioning failures.

3.7 Cartridge functioning.

3.7.1 Cartridge. The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 247 plus or minus (±) 7 feet per second (fps) and a standard deviation not exceeding 4.5 pfs.

3.7.2 Projectile functioning. The projectile shall function high order as evidenced by fragmentation of the body as well as being capable of defeating mild steel targets.

3.7.3 Safety. There shall be no premature burst (see 6.7)or evidence thereof in any of the test firings.

3.8 X-ray examination of ballistic samples. Prior to forwarding the cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination.

3.9 X-ray fo fuze assemblies prior to assembling to cup and liner assembly. The fuze assembly shall be x-rayed for improperly assembled, missing parts, (i.e. set back pin, set back pin retainer, detent, detent spring) armed or partially armed.

3.10 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and quality assurance provisions Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following type of inspection shall be conducted on this item:

a. Quality Conformance inspection.

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4.3 First article inspection. Not applicable.

4.4 Quality conformance inspection. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain:

a. Cartridge case assembly metal parts from one lot interfix number from one manufacturer.

b. Projectile metal parts from one lot interfix number from one manufacturer.

c. Fuzes from no more than two (2) consecutively produced serial lots from one manufacturer.

d. Primer from one lot number from one manufacturer.

e. Liners and caps from one interfix lot number from one manufacturer.

f. RDX from one lot interfix number from one manufacturer.

g. Loaded spitback assemblies from one lot interfix number from one manufacturer.

h. Propellant from not more than one lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Fuze Assembly		8886349
			NEXT HIGHER ASSEMBLY 8886372

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Location of outside diameter of spitback with fuze outside diameter including perpendicularity with fuze seating surface	200	0.40%	3.2	Gage
102.	Crimp missing or improper		0.40%	3.2	Visual
103.	Pull test of spitback assembly		1.5%	3.2	4.5.3
MINOR					
201.	Distance from fuze face to end of spitback assy		0.65%	3.2	Gage
202.	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886374
4.4.2.2	Cup and Liner Assembly Prior to Loading				NEXT HIGHER ASSEMBLY 8886372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Crack in cup		100%	3.2	Visual
MINOR:	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886374
4.4.2.3	Cup and Liner Assembly				NEXT HIGHER ASSEMBLY 8886372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Cracked cup		100%	3.2	Visual
102.	Liner missing or loose		100%	3.2	Visual
103.	Retaining ring missing		100%	3.2	Visual
104.	Distance from fuze seating surface to top of liner		0.40%	3.2	Gage
105.	Diameter of rotating bands		0.40%	3.2	Gage
106.	Location of bottom diameter of liner with inside diameter of skirt		0.40%	3.2	Gage
107.	Cap missing, loose or torn exposing explosive		0.40%	3.2	Visual
MINOR					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886372
4.4.2.4	Projectile Assembly				NEXT HIGHER ASSEMBLY 8886371
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Weight incorrect		0.40%	3.2	Balance
102.	Torque of crimp of fuze assembly	200	0.65%	3.2	4.5.5
MINOR					
201.	Length from front end to rear of rotating bands		0.65%	3.2	Gage
202.	Crimp improper		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	2.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886371
4.4.2.5	Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Chamber gage failure (after complete cure)		100%	3.2	8827889, 8827890, 8827895
MAJOR					
101.	Security of projectile	80	1.5% *	3.2	4.5.6
102.	Air pressure	200	1.5%	3.2	4.5.7
MINOR					
201.	Total length		0.65%	3.2	Gage
202.	Marking misleading or unidentifiable		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

* In addition, if any cartridge tested has a value of 50 pounds or less or any cartridge or projectile shows any evidence of uncured sealant, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8882362 or 8835105 as applicable
4.4.2.6	Unsealed fiber box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Incorrect number of cartridges		0.40%	3.2	Visual
102.	Separator missing		0.40%	3.2	Visual
MINOR:	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Sealed Fiber Box, Prior to Sealing Bag	1 OF 1	8882362 or 8835105 as applicable
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Heat seal test of packing seals	*	*	3.2	4.5.10
102.	Contained damaged		0.40%	3.2	Visual
103.	Corners not blunted		0.40%	3.2	Visual
MINOR					
201.	Contents loose		0.65%	3.2	Visual
202.	Marking misleading or unidentifiable		0.65%	3.2	Visual

NOTES:
* Sampling and rejection shall be in accordance with MIL-P-116.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8882363 or 8835104 as applicable
4.4.2.8	Unsealed Wooden Packing Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Bag torn or perforated		0.40%	3.2	Visual
102.	Bag improperly sealed		0.40%	3.2	Visual
103.	Marking of bag misleading or unidentifiable		0.40%	3.2	Visual
MINOR:	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50872C(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.9	Sealed Wooden Packing Box		SHEET 1 OF 1		8882363 or 8835104 as applicable
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Box damaged		0.40%	3.2	Visual
MINOR					
201.	Contents loose		0.65%	3.2	Manual
202.	Car seal missing or improperly positioned		0.65%	3.2	Visual
203.	Marking misleading or unidentifiable		0.65%	3.2	Visual

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4.4.3 Testing.

4.4.3.1 Check test for deterioration of primer (See MIL-P-20444). If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-P-20444 immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the cartridge (See 6.11).

4.4.3.2 Holding fixture vent check. The cartridge case holding fixture used in the base plug insertion operation shall have its air vents checked per shift for any observation. If a vent is found to be closed or clogged, the vent shall be cleared and the previous shift's production shall be x-rayed 100 percent to ensure that the propellant charge cups have not been damaged. Cartridge cases found to have damaged charge cups shall be classed defective and removed from the lot.

4.4.3.3 X-ray examination of ballistic samples. Prior to forwarding the cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination. If any critical defect is found, the lot shall be rejected including the ballistic samples. If any major defect is found, it shall be noted and replaced with a good sample and the samples forwarded to the proving ground.

4.4.3.4 X-ray examination of fuze assemblies prior to assembling to cup and liner assembly. Each fuze assembly, prior to assembling to assembly, shall be subjected to this test. If any fuze shows evidence of being improperly assembled, having missing parts, (i.e., set back pin, set back pin spring, detent, detent spring), being armed or partially armed, it shall be rejected.

4.4.3.5 Specific gravity of the charge of cup and liner loading assembly. Two (2) sample cup assemblies shall be selected from each four (4) hour's production and subjected to this test. If any sample fails to comply with the specific gravity requirements specified on the applicable drawing, the four (4) hour's production represented by the sample shall be rejected.

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4.4.3.6 Primer functioning. Sampling shall be in accordance with Paragraph 4.2.3.2.2 of MIL-P-20444B, except that the sample size shall be eight hundred (800) primers from each lot. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected. Primers not originally purchased to this requirement shall be tested for the requirement of Paragraph 3.5.

4.4.3.7 Functioning. Beginning with the first lot produced and continuing until three (3) consecutive lots have been accepted under the criteria specified in 4.4.3.7.1, the Government inspector shall select two hundred and eighty-eight (288) cartridges from each lot for this test. One hundred and forty-four (144) cartridges shall be fired using three (3) randomly selected launchers (forty-eight (48) cartridges per launcher) to impact at approximately 200 meters from the launcher on six inch thick loose gravel pad. The remaining one hundred and forty-four (144) cartridges shall be fired using one (1) launcer to impact at approximately 125 feet from the launcher against a vertical 2 1/2 inch thick mild steel plate. If no low order of failure to defect 2 1/2 inch mild steel malfunction occurs in the first thrity-six (36) cartridges tested, the remaining cartridges may be tested on any thickness steel target (1/2 inch minimum). The plate test samples may be selected prior to sealant cure cycle and tested in increments as the lot is produced for control purpose as well as lot acceptance.

4.4.3.7.1 Rejection. The lot shall be rejected if:

a. Twelve or more fail high order detonation or fail to defeat 2 1/2 mild steel plate upon inital impact.

b. Any primer blows back (as evidenced by perforation of the cup and extrusion of the primer around firing pin causing difficulty in opening weapon). (Major Defect.)

c. Any premature burst occur (See 6.7) (Critical defect).

* d. Mean velocity and/or standard deviation fail to meet the applicable requirement. (Major defect).

e. Any primer misfires (Major defect).

f. (1) Thirty (30) or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major defect).

* NOTE: Velocity shall be measured only on cartridges fired into gravel pad.

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(2) One (1) or more cartridges cases cannot be ejected from the launcher by hand operation of the extractor. (Major Defect).

g. Any projectile sticks in gun bore or fails to attain 60 percent of the mean uncorrected velocity of the lot samples. (A projectile on which a velocity reading was not obtained due to low velocity, shall be considered as a failure). (Major defect).

h. Three (3) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major defect).

i. Any metal part separation occurs prior to impact. (Major defect).

4.4.3.8 Functioning regular production. After three (3) consecutive lots have met the criteria of 4.4.3.7.1, the Government inspector shall select one hundred and forty-four (144) cartridges from each lot for this test. Seventy-two (72) cartridges shall be fired using three (3) randomly selected launchers (twenty-four (24) cartridges per launcher) to impact at approximately 200 meters from the launcher and six inch thick loose gravel pad. The remaining seventy-two (72) cartridges shall be fired using one (1) launcher to impact at approximately 125 feet from the launcher against a vertical 2 1/2 inch thick mild steel place. If no low order of failure occurs in the first thirty-six (36) cartridges tested, the remaining cartridges may be tested on any thickness steel target (1/2 inch minimum).

4.4.3.8.1 Rejection. The lot shall be rejected if:

a. Seven (7) or more fail high-order detonation or proper shape charge functioning at initial impact. (See 6.9) (Major defect).

b. Any primer blows back (as evidence by perforation of the cup and extrusion of the primer cup around firing pin causing difficulty in opening weapon) (Major defect).

c. Any premature burst occurs (See 6.7). (Critical defect)

* d. Mean velocity and/or standard deviation fail to meet the applicable requirement. (Major defect)

* NOTE: Velocity shall be measured only on cartridge fired into gravel pad.

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e. Any primer misfires. (Major defect)

f. (1) Ten (10) or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major defect)

(2) One (1) or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor. (Major defect)

g. Any projectile sticks in gun bore of fails to attain 60 percent of the mean uncorrected velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered as a failure). (Major defect)

h. Three (3) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug on cartridge case base). (Major defect).

i. Any metal part separation occurs prior to impact. (Major defect).

4.4.3.9 Suitability test plan (informational test only). Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) consecutive lots that are accepted in accordance with Paragraph 4.4.3.17, an extra sample of seventy-two (72) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The seventy-two (72) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. 119.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

24 rounds +165 F $\pm$ 5 degrees F.

24 rounds ambient (between +40 and +110 degrees F).

24 rounds -65 degrees F $\pm$ 5 degrees F.

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d. The rounds shall be fired using one (1) launcher, to impact at approximately 125 feet from the launcher against a vertical 2 1/2 inch thick mild steel plate. Test results, failure analysis, reports, and other informaiton, as requested, shall be forwarded to ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S for evaluation.

4.4.3.10 Loading composition. Eight (8) ounces minimum unconsolidated material, shall be selected per shift for this test.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 X-ray examination of ballistic samples. The cartridges, prior to forwarding to the proving ground, shall be subjected to x-ray examination using x-ray equipment or any other method staisfactory to the contracting officer. This is a non-destructive test. Parts so tested may be returned to the lot.

4.5.2 X-ray examination of fuze assemblies prior to assembling to cup and liner assembly. Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by x-ray equipment or any other method satisfactory to the Contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.3 Pull test of spitback assembly from fuze assembly. The fuze assembly shall be placed in an approved fixture and the force specified on the Dwg. shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot. WARNING: This test is conducted on assemblies containing an explosive element. Test barricades (where used), procedures and equipment shall have prior approval.

4.5.4 Specific gravity of charge of cup and liner assembly. The specific gravity of the charge shall be determined by the following method:

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Mark the cup and liner assembly for identification.

Weigh the empty cup and liner assembly and record as "W_e".

Level cup and liner with a leveling device. Fill the cup and liner assembly with water, weigh and record as "W_w.

Pour as much water as possible from cup and liner assembly, dry interior of skirt with a clean cotton cloth or piece of absorbent paper and place assembly, open and down, on absorbent paper. Allow assembly to dry.

Distribute preweighed assemblies among other assemblies and load in regular manner with RDX.

Retrieve the loading assembly and record as "W_l.

Subtract (W - W) to obtain the explosive weight.

Level the assembly with leveling device. Fill assembly with water using care to fill to same point used when filling empty cup and liner with water. (Record as "W_b.)

Empty water from assembly and thoroughly dry surface of liner and assembly interior with a clean cloth or absorbent paper.

Calculate specific gravity from the following equation:

$$\text{Sp. Gr.} = \frac{W_l \qquad W_e}{(W_w - W_e) - (W_b - W_l)}$$

4.5.5 Torque or crimp of fuze assembly on projectile. The projectile shall be placed in an approved fixture and the torque specified on the Dwg. shall applied. This is a non-destructive test. Parts so tested may be returned to the lot.

4.5.6 Security of projectile. The cartridge shall be placed in an approved fixture and the force specified on the Dwg. shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot after undergoing a government approved rework procedure. Stake shall be removed from the loaded case prior to returning to the lot. WARNING: This test is conducted on assemblies containing an explosive element. Test barricades (where used), procedures and equipment shall have prior approval.

4.5.7 Air pressure test of cartridge. The cartridge shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure.

NOTE: Rounds in which it can be proven that the source of leaking is the ogive to fuze crimp are to be considered a "NO TEST" round and shall not be considered a basis for rejecting a lot.

4.5.8 Primer functioning. Each primer of the 800 samples shall be tested as specified in MIL-P-20444B, Paragraph 4.3.2.2.

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4.5.9 Functioning. This test shall be conducted at a Government owned proving ground. The rounds being tested (on the gravel phase only) shall be temperature conditioned at 70°F + 5°F for at least two (2) hours then fired within fifteen (15) minutes after being removed from the temperature soak. Gravel: The cartridge shall be fired to impact on a six inch thick loose gravel pad. The gravel pad used shall have a 1/2 inch maxmimum stone size. Steel: The cartridge shall be fired to impact against a vertical 2 1/2 inch thick sheet of mild steel plate. (Proper safety precautions should be exercise to protect all exposed personnel).

The velocity shall be measured at twenty (20) feet from the muzzle of the launcher, using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. Note: To insure that the chronographs are read properly. The ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more tha plus or minus 0.005 the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation (not more than four (4) rounds in a sample size of 72 or eight (8) rounds in a sample size of 144 may be disregarded). If the applicable number of improper ratios is exceeded, additional rounds must be fired. Observation shall be made and results recorded for functioning, mean velocity and standard deviation. The average of all rounds fired per sample (except above noted improper ratio(s) shall meet the requirements of 3.7.1 (See 6.7)

4.5.9.1 Test validity. If, for any reason, proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.5.10 Heat seal test of packing seals. This test shall be perfomred in accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

4.5.11 Loading composition.

4.5.11.1 Determination of calcium resinate. Transfer an accurately weighed portion of approximately 5 grams (gm) of the sample to a tared medium porosity, sintered glass crucible. Extract the specimen with three separate 20 milliliter (ml) portions of chloroform which have been previously saturated with RDX, maintainig each portion in contact with the residue for approximately five minutes before applying suction. Stir occasionally to break up any lumps and wash

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down the sides of the crucible with each 20 ml portion of chloroform to insure complete removal of the calcium resinate. Aspirate the residue until the odor of chloroform is no longer detectable, place the crucible and contents in a drying oven at 100 degrees Centigrade (°C) to 105°C for one half hour, cool in a desiccator and weigh. Repeat the above extraction, drying and weighing procedures with individual 20 ml portions of chloroform until the loss in weight on subsequent extractions total less than 0.003 gm. Calculate the loss in weight of the crucible plus contents to percent calcium resinate on a moisture-free basis as follows:

$$\text{Percent calcium resinate} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible plus contents
W = weight of specimen on a moisture-free basis

4.5.11.2 Determination of the graphite content. Extract the residue, remaining after calcium resinate (See 4.5.11.1) has been removed with two 20 ml portions of dimethyl formamide, maintaining each portion in contact with the residue for approximately five minutes with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20 ml portions of acetone, maintaining each portion in contact with the residue for approximately one minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml beaker, evaporate to dryness on a steam batch and add several drops of a solution of chromotropic acid in 36 normal (N) sulfuric acid. The formation of a reddish-brown color, as compared to a pink color, obtained by a blank prepared in an identical manner, indicates the presence of RDX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying oven at 100 degrees to 105 degrees C for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue in the crucible to percentage of lubricant in the sample on a moisture-free basis as follows:

$$\text{Percent lubricant} = \frac{100\ A}{W}$$

Where:

A = loss in weight of the crucible plus contents
W = weight of specimen on a moisture-free basis

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4.5.11.3 Determination of RDX. The percent RDX shall be determined by subtracting the percent calcium resinate and lubricant from 100 percent.

4.5.11.4 Determination of moisture content. The moisture content shall be determined using Karl Fischer method in accordance with Standard MIL-STD-650, Method 101.4 with methanol as the special solvent.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The cartridge, shall be packed in accordance with Dwg. 8882362 or 8835105 as applicable.

5.2 Packing.

5.2.1 Level A. The cartridge, preserved and packaged as specified in 5.1.1, shall be packed in accordance with Dwg. 8882363 or 8835104 as applicable.

5.3 Marking. Marking shall be in accordance with Dwgs. 8882362, 8882363, or 8835105, 8835104, and Code of Federal Regulations, Title 49 Parts 100-199.

5.4 Unitization. Unitization shall be as prescribed by the procuring agency.

5.5 Shipping. When cartridges from more than one lot are shipped in one car or trailer, each lot shall be clearly separated, and marked to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. Not applicable.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-I. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

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6.4 Submisson of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S.

6.5 Submission of test data. In, addition to the normal distribution of records, when the cartridges are procured by ARRADCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S.

6.6 Standard deviation shall be calculated from the following formula: (Other approved standard deviation formulas may be used.)

$$S = \sqrt{\frac{\Sigma(X-\bar{X})^2}{n-1}}$$

where: X = each individual value

$\bar{X}$ = sample arithmatic mean $\frac{\Sigma X}{n}$

n = sample size

$(X - \bar{X})$ = the sum of the squares of the differences between the sample mean and each individual value.

6.6.1 For a faster and easier method of arriving at the standard deviation formula, the following alternate method may be used.

$$S = \sqrt{\frac{n \Sigma X^2 - (\Sigma X)^2}{n (n-1)}}$$

6.7 Premature burst. A premature burst is considered to have occurred if the round functions in the gun bore or in flight.

6.8 Combining of proving ground tests. When the contractor for the cartridge is also the contractor for one or more of the components thereof, the proving ground tests of the contractor may be combined with the proving ground tests of the cartridge, to save expenses,.

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6.9 High order functioning or defeat of mild steel plate. Evidence of high order functioning shall be proper fragmentation of the projectile cup, as determined by the proof director. Evidence of defect of mild steel plate shall be through penetration of the plate as determined by the proof director.

6.10 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARRADCOM.

6.11 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies, or to the contractor assembling the primer assemblies into the cartridge; and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

6.12 Proving ground test summary.

TEST	SAMPLE SIZE	REQUIREMENTS
Functioning	First 3 lots - 144	See 3.7, 4.4.3.7, 4.4.3.7.1
	Regular Production - 72	See 3.7, 4.4.3.8, 4.4.3.8.1

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1310-A256

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-50872C Cartridge, 40MM, HEDP, M433 Loading, Assembling and Packing

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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